

SBT-DF203 — Lab 8

DNS Spoofing Forensics — Lab Report

Analyst: Abubakar Mustapha Sadiq

Registration Number: 2025/FWSD/11202

Case ID: SBT-DF203-Lab8-2025FWSD11202-AbubakarMustaphaSadiq

Lab Network: Host-only virtual network — Kali (analyst) and Metasploitable (victim)

1. Executive Summary and Authorization

This report documents a DNS spoofing forensics exercise performed for SBT-DF203 Lab 8, conducted entirely within an isolated, host-only virtual network consisting of a Kali Linux analyst VM and a Metasploitable victim VM. The exercise follows the sequence and terminology of the SBT-DF203 course slides, adapted into a forensic evidence-collection workflow.

Authorization for this exercise is scoped to the trainee's own private virtual lab environment, with no production, campus, or public network involvement at any point.

Scope note: Parts B and C of the lab manual (the instructor-approved controlled DNS/ARP spoofing simulation) require the course-provided `arp.py` and `dns_spoof.py` scripts, executed under instructor supervision. At the time of this submission, neither the script files nor a supervising instructor were made available to the trainee. Consequently, Parts B and C could not be executed, and Parts D and E are presented as a documented methodology rather than as findings from an actual controlled-spoof capture. This is recorded transparently in the relevant sections below rather than fabricated.

2. Lab Network Diagram

The lab network is a single host-only virtual subnet (192.168.181.0/24) with the following hosts:

Role	Hostname	IP Address	MAC Address
Analyst VM	kali	192.168.181.128	00:0c:29:05:04:72
Victim VM	metasploitable	192.168.181.129	00:0c:29:b2:75:c4
Gateway	(VMware host-only gateway)	192.168.181.1 / .2 / .254	00:50:56:c0:00:01

Topology: [Analyst VM: Kali, 192.168.181.128] <---host-only vSwitch---> [Victim VM: Metasploitable, 192.168.181.129], with a shared VMware host-only gateway at 192.168.181.1. No connection exists between this subnet and the internet or any production network, confirmed by an unreachable default-route ping test (see Appendix).

3. Baseline Evidence and Hashes (Part A)

A legitimate, pre-spoofing baseline was captured to establish normal DNS and ARP behavior for later comparison. Because the lab's host-only network has no upstream DNS resolver, a minimal authoritative responder was run on the analyst VM using Scapy to answer queries for the reserved training domain `portal.icdfa.test` — this is ordinary authoritative DNS service, not spoofing, since the analyst VM is the legitimate, declared owner of that name for this lab.

3.1 Training Page

A harmless static warning page (no credential fields) was deployed on the analyst VM via Apache2, confirming the target of the training domain. Its SHA-256 hash is recorded in `reports/training_page_sha256.txt`.

3.2 Baseline Capture

Field	Value
Capture file	evidence/dns_baseline.pcapng

SHA-256	07a3a4059726ea24a41d71e0053783b2c9f073fadc65bbb5efca66583fad708
Capture filter	host 192.168.181.129 and (arp or port 53 or tcp port 80)
Duration	30 seconds

3.3 Legitimate DNS Answer — Recorded Indicators

Indicator	Baseline Value
DNS responder IP	192.168.181.128
DNS responder MAC	00:0c:29:05:04:72
Transaction ID	0xb961
Answer (A record)	192.168.181.128
TTL	300
Query-to-response time	≈ 17.8 ms
Competing answers observed	None — single, clean response

3.4 Captured Packet Summary

```

1 0.000000 192.168.181.129 -> 192.168.181.128 DNS Standard query 0xb961 A portal.icdfa.test
2 0.017823 192.168.181.128 -> 192.168.181.129 DNS Standard query response 0xb961 A
portal.icdfa.test A 192.168.181.128
3 4.996626 192.168.181.129 -> 192.168.181.128 ARP Who has 192.168.181.128? Tell 192.168.181.129
4 4.996673 192.168.181.128 -> 192.168.181.129 ARP 192.168.181.128 is at 00:0c:29:05:04:72
5 5.114677 192.168.181.128 -> 192.168.181.129 ARP Who has 192.168.181.129? Tell 192.168.181.128
6 5.115612 192.168.181.129 -> 192.168.181.128 ARP 192.168.181.129 is at 00:0c:29:b2:75:c4

```

4. Controlled Simulation Method (Part B/C)

Not executed. The course-provided `arp.py` and `dns_spoof.py` scripts referenced by the lab manual were not distributed to the trainee, and no instructor was available to approve and supervise a live ARP/DNS spoofing run, as the manual requires. No spoofing traffic was generated and no such capture exists.

Pre-execution evidence that **was** collected regardless, per the manual's Part B checklist, to document the environment's state prior to any spoofing attempt:

Field	Value
IP forwarding state (before)	net.ipv4.ip_forward = 0
iptables snapshot	Recorded in reports/iptables_before.rules
Direct page reachability test	Confirmed analyst web service reachable by direct IP

Had the instructor-provided scripts been available, the documented method would have been: review target IP, gateway IP, training domain and replacement IP configured in the scripts; confirm the victim can reach the analyst page directly by IP; verify the script targets only the reserved training domain; and run both scripts under a 30-second timeout while a parallel tshark capture recorded ARP, DNS and HTTP traffic between victim and analyst.

5. DNS Anomaly Timeline (Part D)

Not applicable — no controlled-spoof capture exists (see Section 4). The methodology that would have been applied is documented below using the same tshark field extraction validated against the baseline capture in Section 3.

Planned extraction command:

```
tshark -r evidence/dns_spoof_controlled.pcapng -Y 'dns' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
-e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
-e dns.flags.rcode -e dns.a -e dns.resp.ttl
```

Expected anomaly indicators, per the baseline established in Section 3.3: two DNS responses sharing one transaction ID with conflicting answer IPs; a responder MAC that does not match the known legitimate resolver MAC (00:0c:29:05:04:72); and an answer TTL or timing that deviates from the baseline's 300-second TTL and ≈17.8 ms response time.

6. ARP Correlation (Part D)

Not applicable — no controlled-spoof capture exists (see Section 4).

Planned extraction command:

```
tshark -r evidence/dns_spoof_controlled.pcapng -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
-e arp.dst.proto_ipv4 -e arp.dst.hw_mac
```

Expected finding: an ARP reply claiming the gateway or victim IP maps to a MAC other than the addresses recorded as legitimate in Section 3 (gateway 00:50:56:c0:00:01, victim 00:0c:29:b2:75:c4, analyst 00:0c:29:05:04:72).

7. Victim Connection Correlation (Part D)

Not applicable — no controlled-spoof capture exists (see Section 4).

Planned extraction command:

```
tshark -r evidence/dns_spoof_controlled.pcapng -Y 'http.request || tcp.flags.syn==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e http.host -e http.request.uri
```

Expected finding: an HTTP connection from the victim (192.168.181.129) to the analyst VM (192.168.181.128) initiated immediately following the forged DNS answer, consistent with the victim trusting the spoofed response.

8. Baseline-versus-Spoof Comparison (Part E)

The comparison table below is populated only for the baseline column; the controlled-spoof column could not be completed for the reasons stated in Section 4.

Indicator	Baseline (recorded)	Controlled Spoof	Assessment
DNS responder IP	192.168.181.128	N/A	Not captured
DNS responder MAC	00:0c:29:05:04:72	N/A	Not captured
Transaction ID match	0xb961 (single)	N/A	Not captured

Answer IP	192.168.181.128	N/A	Not captured
TTL	300	N/A	Not captured
Response time	≈17.8 ms	N/A	Not captured
Competing answers	None	N/A	Not captured
Gateway ARP mapping	00:50:56:c0:00:01	N/A	Not captured
Subsequent HTTP destination	N/A (baseline only)	N/A	Not captured

9. Alternative Explanations and Confidence Level

No spoofing finding is claimed in this report, so no confidence assessment of a spoofing event is made. For completeness, the alternative explanations that any DNS-anomaly finding must be checked against before concluding spoofing — per the lab manual — are: normal DNS caching, split-horizon DNS, and multiple legitimate answers such as round-robin or CDN responses. None of these apply here since no anomalous capture exists to evaluate.

Confidence level in this report's findings: high for the baseline data presented (directly observed, hashed, and reproducible); **not applicable** for any spoofing determination, since none was made.

10. Cleanup and Defensive Recommendations

No spoofing or ARP-poisoning process was started during this exercise, so no interception cleanup was required. Standard verification was nonetheless performed:

Check	Result
IP forwarding restored/confirmed	net.ipv4.ip_forward = 0 (unchanged)
No arp.py / dns_spoof.py processes running	Confirmed — scripts never existed on this host
ARP table reviewed post-exercise	Recorded in reports/arp_after_cleanup.txt equivalent

Defensive Recommendations

- Monitor DNS responses arriving from unexpected MAC addresses or hosts.
- Detect multiple answers for the same DNS transaction ID with conflicting IPs.
- Use DNSSEC validation where supported, and encrypted/authenticated DNS transport where appropriate.
- Deploy Dynamic ARP Inspection, DHCP snooping, switch port security, and network segmentation.
- Enforce HTTPS certificate validation — a forged DNS answer should not be able to produce a valid certificate for the real service.
- Correlate DNS, ARP, endpoint cache, switch, web proxy, and TLS evidence together rather than any single source.

11. Appendix

A. Interface Configuration (Analyst VM)

eth0: 192.168.181.128/24, MAC 00:0c:29:05:04:72
192.168.181.0/24 dev eth0 proto kernel scope link src 192.168.181.128 metric 100

B. ARP Table (Post-Baseline)

192.168.181.254 dev eth0 lladdr 00:50:56:ef:20:e8 STALE
192.168.181.1 dev eth0 lladdr 00:50:56:c0:00:01 STALE
192.168.181.129 dev eth0 lladdr 00:0c:29:b2:75:c4 STALE

C. Chain of Custody

Field	Entry
Case/lab identifier	SBT-DF203-Lab8-2025FWSD11202-AbubakarMustaphaSadiq
Trainee name	Abubakar Mustapha Sadiq
Registration number	2025/FWSD/11202
Evidence file	evidence/dns_baseline.pcapng
Original SHA-256	07a3a4059726ea24a41d71e0053783b2c9f073fadc65bbb5efca66583fad708
Generation method	Legitimate DNS query captured via tshark during authoritative-responder test
Analysis workstation	Kali Linux VM, eth0 192.168.181.128